# GreenSim: An Open Source Tool for Evaluating the Energy Savings through Resource Dynamic Adaptation

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*Abstract* — **Due to the continuous growth of customer population, the spreading of broadband access, the increase in energy price, and the expanding number of services being offered by telecoms and providers, energy efficiency issue has become a high-priority objective for wired networks and services infrastructures. Current network devices are well-known to be over-dimensioned for rush-hour traffic loads, and to have flat energy requirements with respect to their real utilization. Recent studies suggested to reduce the energy consumption of networks and networked devices through the introduction of adaptive capabilities, which allow to modulate energy requirements according to network-aware performance (i.e., delays, packet losses, etc.). However, at the today's state of the art, the real impact of these power management techniques is not clear, and requires further studies and researches. Based on this context, we focus on development of a new simulator called "GreenSim", that aims at analyzing and evaluating the impact of adaptive energy saving techniques both on network- and energy-sides in the presence of real traffic traces and data. The results achieved with the proposed simulator demonstrate that these green technologies, when applied to packet processing engines inside network devices, can save up more that 40% of energy consumption in the presence of real Internet traffic profiles, while maintaining an acceptable network performance level.**

***Keywords: green simulator, simulation, performance evaluation, networking, packet processing engine, burst analysis.***

## I. Introduction

Energy Efficiency is one of the key aspects of the Future Internet that should pervade the network infrastructure as a whole, to such extend as to become a part of the network design criteria [2].

The Information and Communication Technologies (ICT) field has been reasonably considered as a key factor to develop new services and systems, with the intent to reduce and/or monitoring energy wastes and achieve high levels of efficiency [2]. In such context, different projects are developed in order to build "energy aware" environments.

It is awkward that, when the role of ICT is becoming more important in addressing better energy efficiency in the energy distribution and production sector, ICT would not apply the same concepts to itself [3].

Only recently, due to the raise in energy price, the continuous growth of customer population, the increase in broadband access demand, and the expanding number of services being offered by telecoms and providers, the energy efficiency has become a high-priority objective for wired networks and service infrastructures.

These rising trends in network energy consumption essentially depend on new services offered which follows Moore's law, by doubling every two years. In order to support new generation network infrastructure and related services, telecoms and service providers need a larger number of devices, with sophisticated architectures able to perform more and more complex operations in a scalable way. Notwithstanding these trends, it is well known that networks, links and equipment are provisioned for busy or rush hour load, which typically exceeds their average utilization by a wide margin. While this margin is generally filled in rare short time periods, the overall power consumption in today's networks remains more or less constant with respect to different traffic utilization levels.

There are two main motivations the drive the quest for "green" networking:

a) environmental one, connected to the reduction of wastes and relative $CO_2$ emission;

b) economic one, related to the reduction of costs sustained by operator to keep network on and maintain the desired level of quality of service.

In the last few years, the telecoms around the world reported statistics of network energy requirements and the related carbon footprint, showing an alarming and growing trend. In 2005 the European Commission DG INFSO report in [4] estimated European telcos and operators to have an overall network energy requirement equal to 14.2 TWh which will raise to 35.8 TWh in 2020 if no green network technologies will be adopted.

To this purpose telecoms and service providers have begun requiring disruptive architectural solutions, protocols and innovative equipment in order to allow a better ration of performance to energy consumption. This has inspired major

ICT companies and research bodies to undertake different private initiatives focused on developing more sustainable data centres and network infrastructures.

Nevertheless the overall power consumption in today's networks remains more or less constant with respect to different network utilization levels. The key of any advanced power saving criteria certainly resides in dynamically adopting resources, provided at network, link or equipment levels, to current traffic requirements and loads.

In this respect, current green networking approaches range from novel traffic engineering and routing criteria [6], [7], to the introduction of energy-aware equipment [8], components [9] and network interfaces [10].

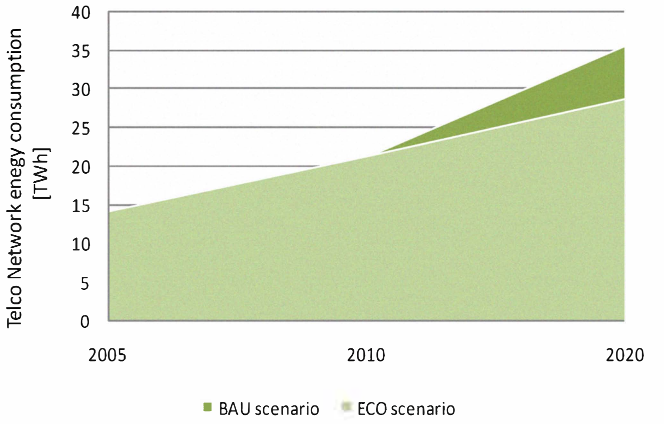


Figure 1. Energy comsumption estimation for the European telcos' network infrastructures in the "Business-As-Usual" (BAU) and in the Eco sustainable (ECO) scenarios. Source: European Commission DG INFSO report in [4].

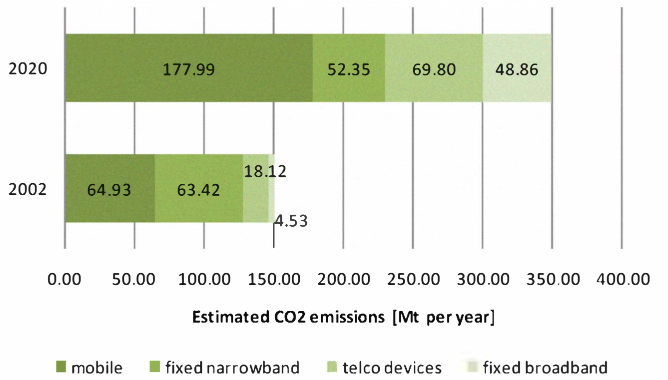


Figure 2. Greenhouse gas emission estimation according to GeSI [5].

A new energy efficient version of the Ethernet standard, namely 802.3az, is described in [11] and [12]. This version is based on the Adaptive Link Rate (ALR) concept. ALR is realized in order to adjust network interface speed and consequently power to effective traffic levels, and when no traffic is crossing the link.

In this work, we started from similar concepts with respect to the ALR, which allow us to trade power consumption for network performance. However, considering the well-known fact [8] that power consumption in today's network equipment mainly arises more on device internal chips (especially the ones that realize the data-plane functionalities), than on network interfaces, we especially focus on how using green adaptive techniques for reducing energy consumption and maintaining a given level of network performance.

In this respect, the first main objective of this paper is to provide and evaluate a new network device simulator in order to compare power consumption of a single forwarding network node. The evaluation is made with real internet traffic. Our idea is to analyze how to save up energy adopting different green technologies that today are already available on Components-Off-The-Shelves (COTS) processors and under development in other hardware technologies (e.g., network processors, ASIC [13] and FPGA [14]).

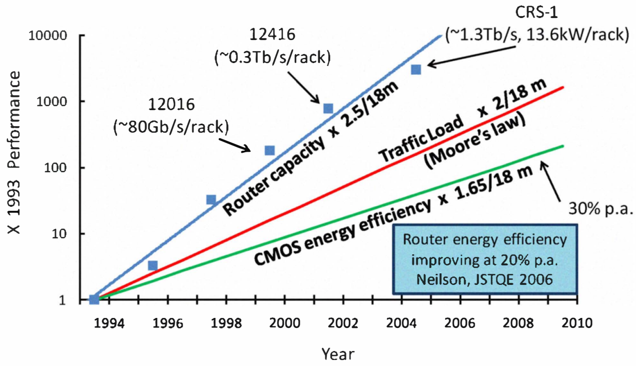


Figure 3. Evolution from 1993 to 2010 of routers capacity vs. traffic volumes (Moore's law) and energy efficiency in silicon technologies. SOURCE: G. Epps, Cisco Systems, 2006.

In few words and as described in section I in a more detailed way, these capabilities allow:

a) minimizing power consumption when no activities are performed (known as C-State optimization);

b) modifying the trade-off between performance and energy consumption where the hardware is active and performing operations (P-State optimization);

At lowest level, these mechanisms are generally realized at the hardware level, by powering off sub-components, or by changing the operating frequency and called through the ACPI (Advanced Configuration and Power Interface) standard.

The proposed "GreenSim" simulator allows to determine the impact of using these adaptive mechanisms in the network context by evaluating both the potential energy savings and the network performance. Our simulations were done with real Internet traffic data to individuate the advantages using energy-aware capabilities. The results show that it is possible with the proposed model save up more than 40% of energy respect the case without any optimizations.

This paper is organized as follows. The two power management techniques, namely C- and P- states, available in legacy hardware through the ACPI standard, are introduced in section II. In section III is described the "GreenSim" simulator and the model used to develop it. Section IV reports numerical results of the simulation with real Internet traffic data and the difference of performance and power consumption with and without the "green approach". Finally, in section V are described the conclusions and the related future works.

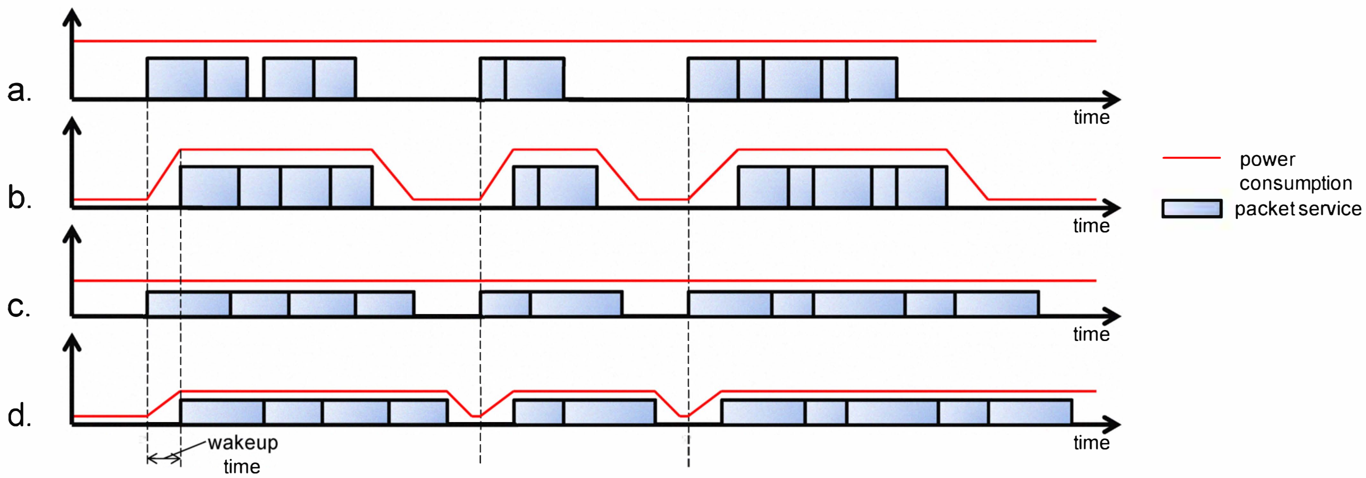


Figure 4. Power consumptions in the following cases: (a) no power-aware optimizations, (b) only idle logic, (c) only performance scaling, (d) performance scaling and idle logic.

## I. Network Performance and Energy Aware Adaptation

Nowadays, the largest part of current network equipment does not include power management capabilities. As a consequence of this absence, power management is a key factor in today's processor across all market segments.

The ACPI provides a well-known standardized interface between the hardware and the software layers in order to hide the heterogeneous power management mechanism of different processors.

They use several techniques to reduce their energy usage by exploiting the fact that the systems do not need to run at peak performance all the time.

These different techniques are obtained by tuning the clock frequency and/or voltage of processors, or by throttling the CPU clock (i.e. the clock signal is disabled for some number of cycles at regular intervals). Decreasing the operating frequency and the voltage of the processors, or throttling its clock, obviously allows the reduction of the power consumption and of head dissipation at the price of slower performance.

The ACPI standard introduces two main different power saving mechanisms, namely performance and power status (P-States, and C-States respectively).

TABLE I. Energy Saving and Transition Times for COTS Processors' C-States

| *C-State* | *Energy Saving with respect to $C_0$ state* | *Transition Times* |
|---|---|---|
| $C_1$ | 70% | 10 ns |
| $C_2$ | 75% | 100 ns |
| $C_3$ | 80% | 50 µs |
| $C_4$ | 98% | 160 µs |
| $C_5$ | 99% | 200 µs |
| $C_6$ | 99.9% | unknown |

Regarding the C-states, the $C_0$ power state is an active state where CPU executes instructions, while the power states from $C_1$ to $C_n$ are idle states where the processor consumes less power and dissipates less heat. As the sleeping state becomes deeper the transition between the active and the sleeping state (and vice versa) requires longer time. The number of C-states *n* depends of the processors implementation. Current COTS processors generally includes up to 6 C-states.

ACPI allows the tuning of performance thought P-states transitions. They consent to modify the operating energy by altering the working frequency and/or voltage, or throttling its clock. Thus using P-states, a core can consume different amounts of power while providing different performance at the $C_0$ (running state). In general, the higher the index of P and C states is, the less will be the power consumed, and the heat dissipated.

Unfortunately, due to issues in silicon electrical stability, the transition time between different P-states is generally very slow especially if compared with usual time scales in network dynamics. The large part of current CPUs can switch their performance state in about 100ms. Due to these large P-state transition times, any closed-loop policies with tight time constraints are not feasible and cannot be adopted for optimizing power consumption inside network device architectures.

In addition ACPI standard provides specific control applications, namely governors, which are devoted to dynamically configure power profiles in terms of C- and P-state through the ACPI standard interfaces.

The features described above are available in normal COTS processors. These techniques are developed to save energy in personal computer used for general purpose. No specific technique was developed to reduce the power consumption in network devices.

The most important techniques to save energy in network devices are, as described above, substantially two: dynamic frequency scaling to adapt the power consumption to the traffic volumes, and minimizing power consumption when no activities are performed. In Figure 4 are showed the different cases using these power saving techniques. Reduce the power consumption with performance scaling and idle logic during no activities can be very difficult when this period is small (sub-figure d).

## II. THE GREENSIM ARCHITECTURE

As showed in Figure 4 there are two different logic to save energy. We developed a network device simulator "GreenSim" [1] to evaluate the impact of these energy-aware techniques on the performances.

At the high level, the GreenSim is able to capture the working dynamics of a packet processing engine by mainly considering two different periods as well as the transition between them:

a) *on* period, where the Packet Processing Engine executes instructions (in this case, forwards packets) and it has the pair of ACPI state set to $C_0$ and $P_0$.

b) *off* period, where the Packet Processing Engine does not forward any packets and consumes less power energy. It can be set in one of $C_1 - C_n$ ACPI states.

Our approach considers a network device substantially as a traffic forwarding engine that processes incoming bursts of packets. A finite buffer, with a size of equal to *N* bursts, is assumed to be bound to the device for backlogging incoming traffic bursts.

The main idea is set the device in *off* period when there are no incoming bursts in the queue to process. As the transition time of the C-states where the network device switch from *on* to *off* period (and vice versa) there are additional transition times can allow the change of the state. So the transition times between these two states are not instantaneous, and a transition time $\tau_t$ is required.

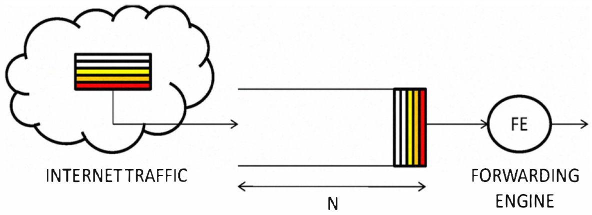


Figure 5. Overview of network device model

When a new burst is received, the device has to wake up by exiting from the *off* period, and returning to the active *on* period and this requires an additional $\tau_s$ period.

Furthermore the time $\tau_s$ is required to suitable configure the elaboration process. For instance, as shown in [15], in Linux based SW routers, $\tau_s$ represents the computation time of HW interrupts that are used to trigger the packet forwarding process.

We can define a period named $T_P^{(n)}$ which identifies the classic "cycles" of form:

$$T_P^{(n)} = T_{ON}^{(n)} + T_{OFF}^{(n)} \tag{1}$$

where $T_{ON}^{(n)}$ is the *n*-th period in which the device is in the *on* period while $T_{OFF}^{(n)}$ is the *n*-th period in which the device is in the *off* period.

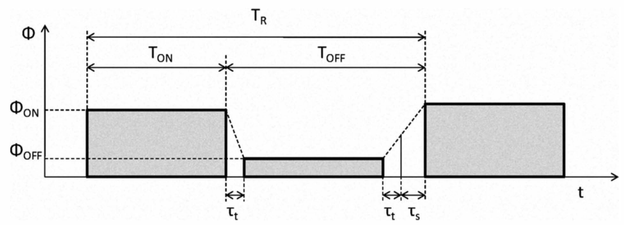


Figure 6. Power consumption during a renewal busy idle cycle

Thus, during a generic $T_P^{(n)}$, the instantaneous power requirement will be equal to $\Phi_{off}$ only if and when the device resides in the *off* period. On the contrary during *off-on* transitions and setup times the power consumption is supposed to be $\Phi_{on}$. Thus it is possible to express the energy wasted during a generic $T_{OFF}^{(n)}$ as follows:

$$E_{OFF}(T_{OFF}^{(n)}) = \begin{cases} \Phi_{ON} T_{OFF}^{(n)}, & T_{OFF}^{(n)} < \theta \\ \Phi_{ON}\theta + (T_{OFF}^{(n)} - \theta)\Phi_{OFF}, & T_{OFF}^{(n)} \geq \theta \end{cases} \tag{2}$$

where $\theta = 2\tau_t + \tau_s$.

Thus, we can obtain the total energy wasted during idle periods by the sum of every $E_{OFF}(T_{OFF}^{(n)})$:

$$E_{OFF} = \sum_{n=1}^{T} E_{OFF}(T_{OFF}^{(n)}) \tag{3}$$

where *T* is the total number of periods.

The proposed simulator computes the total power consumption considering the specific traffic data as inputs. In addition it obtains other important measurements:

a) $\bar{L}$, average latency

b) $\lambda$, average burst inter-arrival time

c) $\mu$, effective throughput

With these measurements it is possible to understand how the different "green capabilities" can decrease the performances.

## III. NUMERICAL RESULTS

Regarding numerical results, we consider a device with power management capabilities similar to the one of the Linux SW Router used in [15]. This choice is mainly due to the fact the current HW routers do not include power management features, and only their nominal and/or maximum power consumptions are reported in the datasheets. Tables II and III report the usual parameters for the ACPI states.

TABLE II. POWER CONSUMPTIONS, TRANSITION TIMES OF THE DEVICE'S C-STATES

| *C-State* | $\Phi$ | $\tau_t$ |
|---|---|---|
| $C_1$ | 10 W | 10 ns |
| $C_2$ | 8 W | 100 ns |
| $C_3$ | 7 W | 50 µs |

TABLE III. POWER CONSUMPTIONS, FORWARDING CAPABILITIES OF THE DEVICE'S P-STATES

| *P-State* | *Φ* | *μ* |
|---|---|---|
| $P_4$ | 50 W | 650 kpkts/s |
| $P_3$ | 60 W | 770 kpkts/s |
| $P_1$ | 70 W | 890 kpkts/s |
| $P_0$ | 80 W | 1010 kpkts/s |

We estimated the $\Phi_{ON}$ and $\Phi_{OFF}$ with the values reported in Tables II and III. For the *on* period we consider to run on the maximum P-state ($P_0$) and C-State ($C_0$). For the off period we consider the $C_1$ state with 10 W of power consumption. Table IV show these assignments.

The traffic traces used in our experiments are real traffic profiles, captured by the MAWI Working Group and publicly available at the URL in [16]. In detail, the used traffic trace is long 96 hour, and were obtained at March 2009 on a trans-continental link between Japan and USA with a 150Mbps speed. It was also adopted in the a "Day in the Life of the Internet" project [17].

TABLE IV. POWER CONSUMPTION AND TRANSITION TIMES OF *ON* AND *OFF* PERIODS.

| | *Φ* | $\tau_t$ |
|---|---|---|
| $\Phi_{ON}$ | 80 W | - |
| $\Phi_{OFF}$ | 10 W | 10 ns |

To obtain a more intuitive analysis it was divided the traffic data into time-slices of 15 minutes. For every time-slice are computed the measures about the number of burst and packets, the average number or packets per burst and the average burst inter-arrival time. These stats are showed in Figures 7, 8, and 9, respectively.

The traffic data are characterized by two periods of low traffic volume and a central period of intensive traffic.

The proposed simulator computed the amount of power consumption using the model described above and with the value of energy consumption showed in Table IV.

The parameters of power consuming used in this simulation are very conservative. Actually the *off* period has the values of ACPI $C_1$ state ($\Phi$ e $\tau_t$).

Nevertheless, the power saving that achieves from the simulator is satisfactory. In Figure 10 it is showed the power consumption of the different time slice. The power saving adopting this simple "green technique" is high.

The accurate amount of power saving can be computed with the following formula:

$$\Phi_{ON}T - (E_{ON} + E_{OFF}) \tag{4}$$

where $E_{OFF}$ is the total power consumption when the device is in *off* period and it is defined in (2) and (3). Instead $E_{ON}$ can be expressed as:

$$E_{ON} = E_{ON}(T_{ON}^{(n)})T_{ON} \tag{5}$$

where $T_{ON}$ is the total number of period in which the machine is in the *on* period.

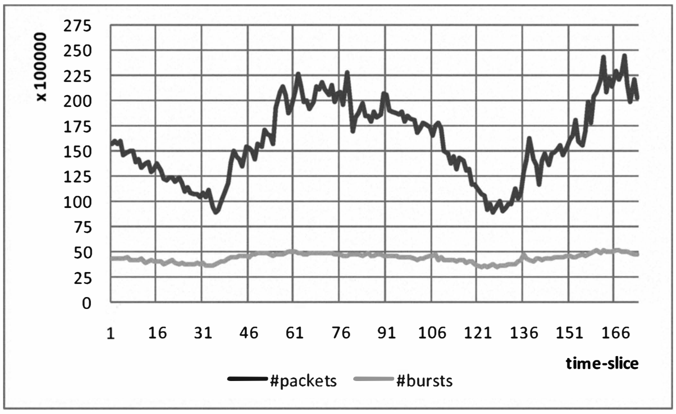


Figure 7. Traffic Data: number of bursts and packets per time-slice.

In Figure 10 it is very simple to individuate that the power saving is more than 40%. The performances with the proposed model showed in Figures 10, 11, and 12 are not significantly worsted. The size of the queue is nearly always zero (the results about this statistical parameters are not so important to show in a graph). These results confirm that the simple technique to optimize the power consumption only with two period (*on*, *off*) allow to obtain a significant power saving without compromise the performances.

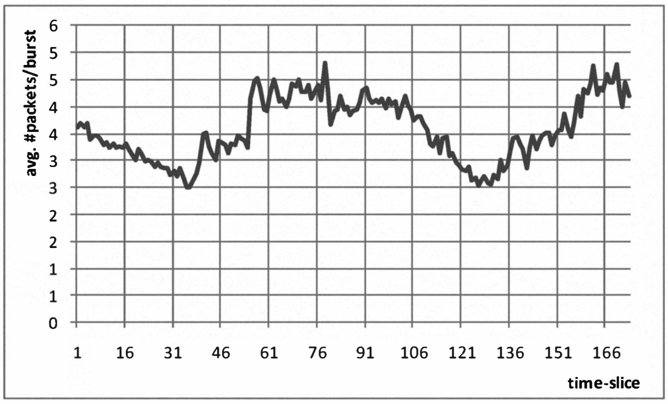


Figure 8. Traffic Data: average number of packets per burst.

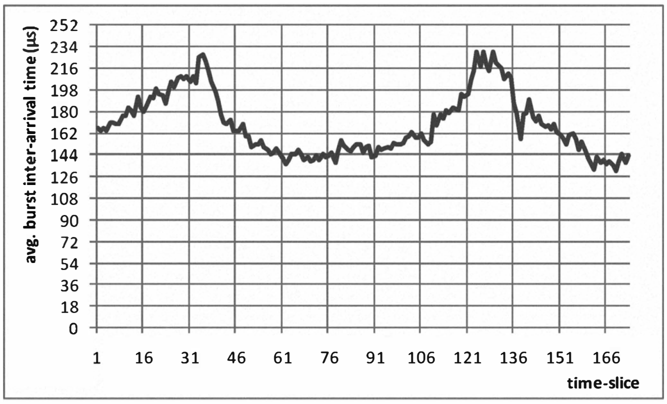


Figure 9. Traffic Data, average burst inter-arrival time.

Figures 13 and 14 report the energy saving and the network performance, respectively, which can be obtained by using the C2 and the C3 states with respect to the C1 one. In more detail, observing Figure 14, we can outline how the adoption of the C3 state does not provide additional energy savings with respect to the C2 ones. This effect is substantially due to the C3 transition time value, which is too large with respect to packet and burst level dynamics, and it consequently prevents the packet processing engine to effectively and fully enter in the low power standby state.

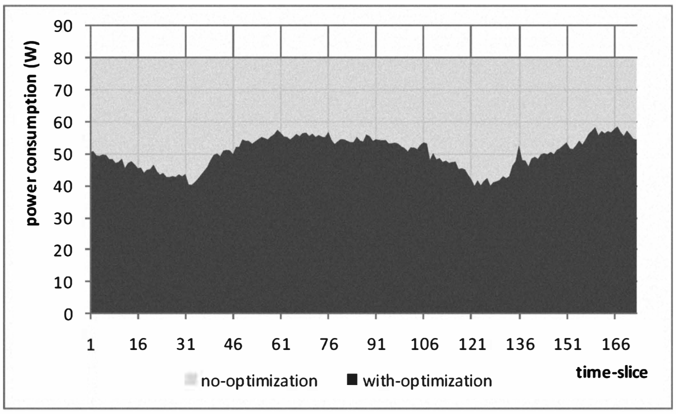


Figure 10. Comparison betweeen power consumption with the proposed"green model" and the standard power consumption without power saving capabilities.

Moreover, as shown in Figure 14 the adoption of C3 state causes very large performance decay in terms of packet forwarding latency. In detail, while the latency in the C2 state differs on the average of only 0.52μs with respect to C2 case, the C3 state lead to average values larger that 10μs than the previous cases.

## IV. Conclusion and Future Works

In this paper we focused on performance-adaptive network devices, able to save energy by scaling their capacities. In detail, we proposed a simulator, called "Green Simulator", in order to analyze the impact of power management capabilities on network performance metrics. The simulations were done with real internet traffic with a simple model developed to save energy without compromise the performance constraints. The saved energy is more than 40% of the power consumption without optimization. The results of these simulations showed that an adaptive approach to save energy is necessary and possible with real internet traffic. With the described model the constraints about latency and throughput are satisfied. For these reasons this contribution can be considered as an important starting point to build and design optimized next-generation green devices. The future works aim to extend the analysis to the network interface and study the impact of power consumption in a chain of network devices.

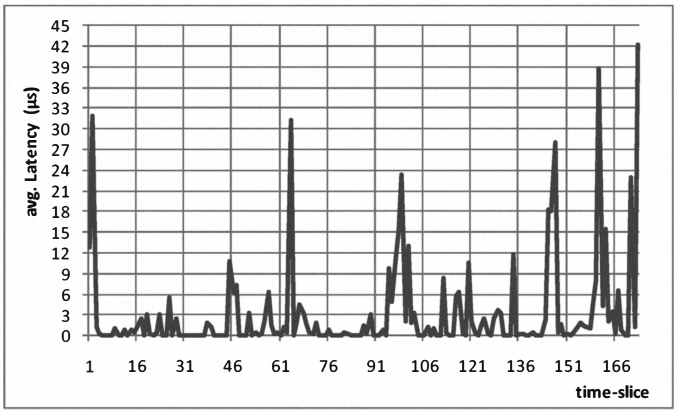


Figure 11. Average Latency (in μs) for each time-slice.

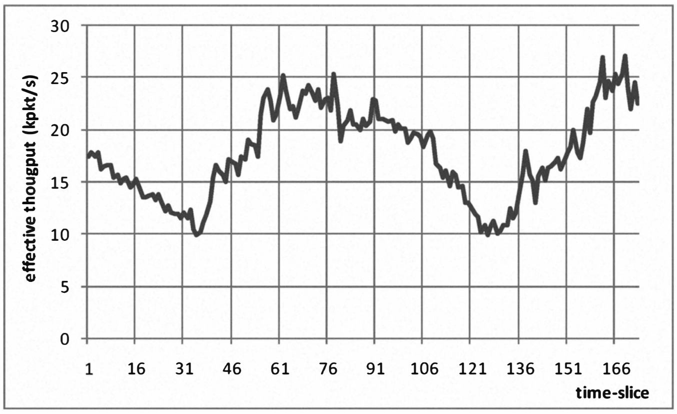


Figure 12. Effective throughput (in Kpkt/s) for each time-slice.

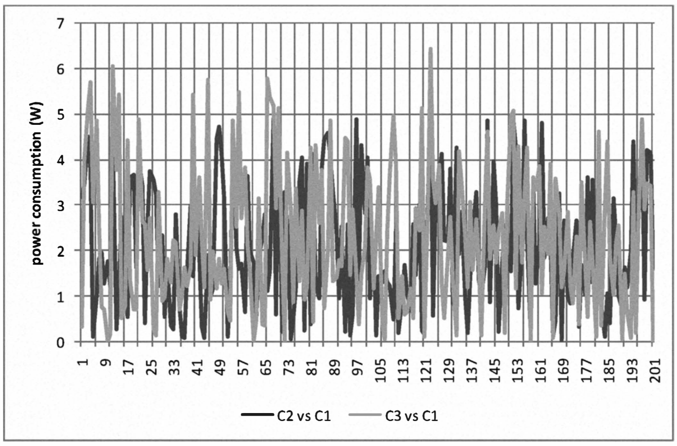


Figure 13. Additional energy savings when the C2 and C3 states are adopted. The values are obtained by taking the C1 case as term of comparison.

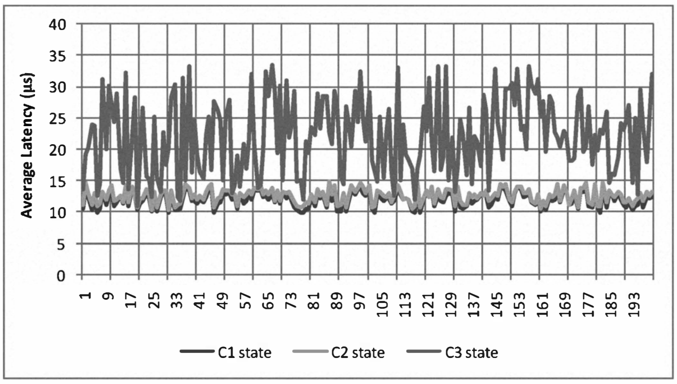


Figure 14. Packet latency values according to different C-states.

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